

■ BusinessNext

Cost Estimator

Pricing & Infrastructure Report

BOI

Mode: SaaS — BusinessNext
Hosted

Date: 09 June 2026

Inflation: 4% / yr

AWS Region: us-east-1

Executive Summary

High-level cost overview and infrastructure snapshot

\$12,390	\$148,685	\$805,325	15,500
Monthly Cost	Annual Cost	5-Year Total	Named Users
Production + environments	Year 1 total	At 4% inflation/yr	Year 1 licensed

Monthly	Annual	5-Year
\$12,390	\$148,685	\$805,325

Project Details

Customer / Bank	BOI
Client Mode	SaaS — BusinessNext Hosted
Database Type	PostgreSQL
Total Named Users	15,500
Concurrent Users	4,650 (30% of named)
Mobile Users	4,050
Worker Nodes	9
Total vCPUs (worker)	96
Total RAM GB (worker)	172 GB
Data Size	2100 GB
Estimate Date	09 June 2026

Node Distribution

AI-assisted infrastructure architecture

Total Worker Nodes	9
Total DB Nodes	11
Fixed / Managed Roles	7
ClickHouse OLAP Nodes	5
Confidence	High

Category	Role / Service	Nodes	Instance Fam.	vCPU/node	RAM GB/node	Model
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	Compute Intensive	8	16	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	Compute Intensive	8	16	Reserved 3 Yr
PGSQL Database	Linux Nodes – Database server (Primary)	2	Memory Intensive	32	128	None
PGSQL Database	Nodes – Cluster (etcd+haproxy, pgbackrest)	4	Memory Intensive	2	8	None
PGSQL Database	Storage: SAN (Primary PostgreSQL DB)	2	10K IOPS	—	—	None
S3	S3 Storage + Replication	1	Cloud Storage Service	—	—	On Demand
Security & Compliance	WAF + AWS Shield Advanced	1	Managed Service	—	—	On Demand
Security & Compliance	CloudTrail + Config Audit Logs	1	Managed Service	—	—	On Demand
Kubernetes Cluster	Cloud Managed K8s	1	K8s Service	—	—	None
Caching & Session	Elasticaache Service	2	Memory Optimized	4	16	Reserved 3 Yr
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	—	—	On Demand
Infrastructure & Monitoring	Bastion Host	1	General Purpose	2	4	Reserved 3 Yr
Infrastructure & Monitoring	NAT Gateway	2	NAT	—	—	On Demand
Infrastructure & Monitoring	Image Registry (ECR)	1	Managed Service	—	—	On Demand
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage Service	—	—	On Demand

ClickHouse OLAP Cluster

Self-hosted sharded/replicated ClickHouse DB + Keeper coordination cluster

Analytics Data Volume	4,284 GB (2.0× transactional DB × 1.02× volume factor)
Total Cluster Nodes	5 (2 DB + 3 Keeper)
DB Shards × Replicas	1 shard(s) × 2 replica(s)
Total vCPU	28
Total RAM	112 GB
Total Storage	12,600 GB (EBS io2 + gp3)
DB Node Spec	8 vCPU / 32 GB RAM / 6,000 GB SSD each
Keeper Node Spec	4 vCPU / 16 GB RAM / 200 GB SSD each
Hosting	Self-hosted on EC2 — same for SaaS and On-Prem

ClickHouse DB Cluster — Sharded + Replicated

Shard	Node	vCPU	RAM (GB)	Total RAM/Shard (GB)	Total Cores/Shard	Storage/Node
Shard 1	Replica 1	8	32	64	16	6,000 GB
Shard 1	Replica 2	8	32	64	16	6,000 GB

ClickHouse Keeper Cluster — 3-Node Raft Quorum

Type	Node	vCPU	RAM (GB)	Total RAM (GB)	Total Cores	Storage/Node
Keeper Node	Node 1	4	16	48	12	200 GB
Keeper Node	Node 2	4	16	48	12	200 GB
Keeper Node	Node 3	4	16	48	12	200 GB

AI Services Infrastructure

Predictive AI · GenAI · Agentic AI — Kubernetes cluster sizing (Maybank template)

AI Types Enabled	Predictive AI · GenAI · Agentic AI
Total AI Worker Nodes	13
Total AI vCPU	192
Total AI RAM	384 GB
Total AI Storage	10,116 GB
AWS Bedrock (LLM API)	\$3,000/month — No self-hosted GPU

AI Cluster Configuration (Production)

AI Type	Nodes	Total Cores	Total RAM	Total SSD	Databases	Est. Compute/mo
Predictive AI	2	16	32 GB	500 GB	—	\$841/mo
GenAI	2	32	64 GB	400 GB	CRM Database (SQL DB) (500 GB)	\$841/mo
Agentic AI	9	144	288 GB	9,216 GB	Vector DB (Milvus) (200 GB); CRM Database (SQL DB) (800 GB)	\$3,784/mo

GPU Requirement	None — all LLM inference via AWS Bedrock (managed)
Bedrock API Cost	\$3,000/month (user-configured, token-based)
Annual Bedrock	\$36,000
5-Year Bedrock	\$202,787 (at 4% inflation)

AWS Cost Breakdown

Service-by-service monthly and annual costs

Compute & Database Roles

Category	Role	Nodes	Instance	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	c6i.4xlarge	\$0.68	\$3,101	\$37,215
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	c6i.2xlarge	\$0.34	\$289	\$3,470
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	c6i.2xlarge	\$0.34	\$660	\$7,923
PGSQL Database	Linux Nodes – Database server (Primary)	2	r5.8xlarge	\$2.02	\$2,991	\$35,896
PGSQL Database	Nodes – Cluster (etcd+haproxy, pgbackrest)	4	r5.large	\$0.13	\$400	\$4,799
PGSQL Database	Storage: SAN (Primary PostgreSQL DB)	2	EBS io2	\$0.00	\$525	\$6,300
S3	S3 Storage + Replication	1	EBS io2	\$0.00	\$738	\$8,850
Kubernetes Cluster	Cloud Managed K8s	1	EKS	\$0.10	\$73	\$876
Caching & Session	ElastiCache Service	2	cache.r6g.large	\$0.33	\$482	\$5,782
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	\$0.00	\$33	\$394
Infrastructure & Monitoring	Bastion Host	1	c6i.large	\$0.09	\$86	\$1,033
Infrastructure & Monitoring	NAT Gateway	2	NAT	\$0.00	\$41	\$497
Infrastructure & Monitoring	Image Registry (ECR)	1	ECR	\$0.00	\$51	\$614
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	S3	\$0.00	\$118	\$1,413
ClickHouse DB Cluster	ClickHouse DB — Shard 1, Node 1	1	r5.2xlarge	\$0.50	\$1,118	\$13,415
ClickHouse DB Cluster	ClickHouse DB — Shard 1, Node 2	1	r5.2xlarge	\$0.50	\$1,118	\$13,415
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 1	1	m5.xlarge	\$0.19	\$156	\$1,874
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 2	1	m5.xlarge	\$0.19	\$156	\$1,874
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 3	1	m5.xlarge	\$0.19	\$156	\$1,874
Operations	CloudWatch Monitoring	0	CloudWatch	\$0.00	\$98	\$1,170

Cost Summary

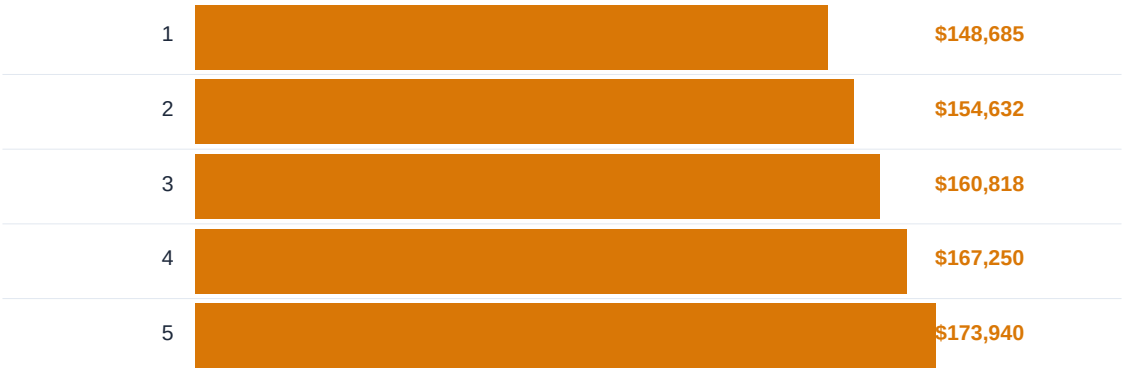
Total Monthly (Production)	\$12,390
Total Annual (Year 1)	\$148,685
5-Year Total (with inflation)	\$805,325
Inflation Rate Applied	4%

5-Year Inflation Forecast

Cost projection at 4% annual inflation

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative Total
Year 1	1.0000×	\$12,390	\$148,685	\$148,685
Year 2	1.0400×	\$12,886	\$154,632	\$303,317
Year 3	1.0816×	\$13,401	\$160,818	\$464,135
Year 4	1.1249×	\$13,938	\$167,250	\$631,385
Year 5	1.1699×	\$14,495	\$173,940	\$805,325

Annual Cost Trend



Total 5-Year Investment	\$805,325
Average Annual Cost	\$161,065
Average Monthly Cost	\$13,422
Year-over-Year Growth	4.0% (inflation) applied from Year 2

Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

Environments: Pre-Prod, SIT, UAT

Environments Selected	3 — Pre-Prod, SIT, UAT
Base Cost per Env / mo	\$4,956
Base Cost per Env / yr	\$59,468
Calculation	3 envs × \$4,956/mo = \$14,867/mo total

Monthly	Annual	5-Year
\$14,867	\$178,404	\$0

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (Pre-Prod)	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – Pre-Prod	4	c6i.4xlarge	\$2,082
Kubernetes	Worker Nodes (Monitoring/EFK) – Pre-Prod	1	c6i.2xlarge	\$289
PGSQL DB	DB Server (Primary) – Pre-Prod	1	m5.8xlarge	\$1,145
PGSQL DB	Storage: SAN – Pre-Prod	1	EBS io2	\$105
S3	S3 Storage + Replication – Pre-Prod	1	S3	\$54
Caching	ElastiCache – Pre-Prod	1	cache.r6g.large	\$121
Infrastructure	Load Balancer – Pre-Prod	1	ALB	\$16
Infrastructure	NAT Gateway – Pre-Prod	2	NAT	\$32
Infrastructure	Bastion Host – Pre-Prod	1	t3.medium	\$54
Infrastructure	Image Registry (ECR) – Pre-Prod	1	ECR	\$51
Backup	Back Up – DB & Infra Logs – Pre-Prod	1	S3	\$74
Operations	CloudWatch Monitoring – Pre-Prod	0	CloudWatch	\$34
ClickHouse DB Cluster	ClickHouse DB Nodes (1s×1r) – Pre-Prod	1	r5.2xlarge	\$668
ClickHouse Keeper Cluster	ClickHouse Keeper Nodes (1) – Pre-Prod	1	m5.xlarge	\$156

Disaster Recovery (DR) Environment

Monthly	Annual	5-Year
\$13,769	\$165,222	\$930,692

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (DR)	1	EKS	\$73

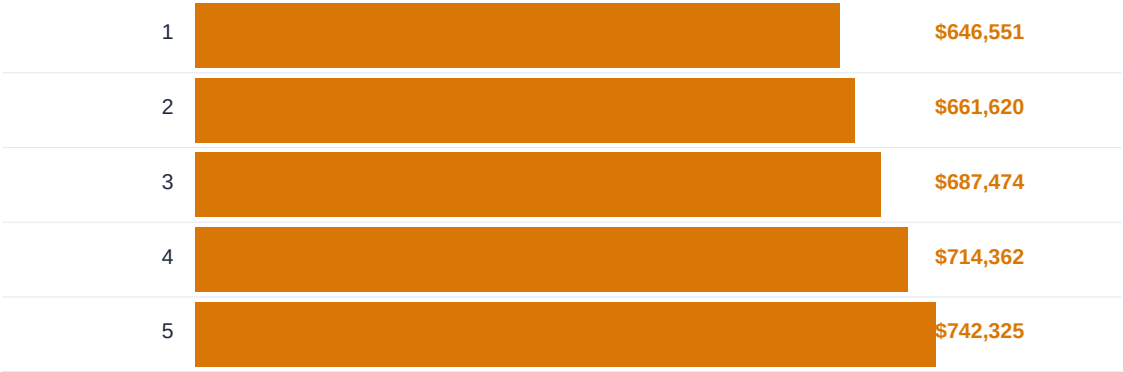
Category	Role	Nodes	Instance	\$/month
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – DR	9	c6i.4xlarge	\$4,560
Kubernetes	Worker Nodes (Monitoring/EFK) – DR	2	c6i.2xlarge	\$578
PGSQL DB	DB Server (Primary) – DR	2	m5.16xlarge	\$4,533
PGSQL DB	Storage: SAN – DR	2	EBS io2	\$525
S3	S3 Storage + Replication – DR	1	S3	\$136
Caching	ElastiCache – DR	2	cache.r6g.large	\$242
Infrastructure	Load Balancer – DR	2	ALB	\$33
Infrastructure	NAT Gateway – DR	2	NAT	\$32
Infrastructure	Bastion Host – DR	1	t3.medium	\$54
Infrastructure	Image Registry (ECR) – DR	1	ECR	\$51
Backup	Back Up – DB & Infra Logs – DR	1	S3	\$184
Operations	CloudWatch Monitoring – DR	0	CloudWatch	\$62
ClickHouse DB Cluster	ClickHouse DB Nodes (1s×2r) – DR	2	r5.2xlarge	\$2,236
ClickHouse Keeper Cluster	ClickHouse Keeper Nodes (3) – DR	3	m5.xlarge	\$468

PUPM Analysis

Price Per User Per Month — 5-year projection

Year	Named Users	Monthly Usage	Platform Annual	AWS Total	Discounted Cost	PUPM
Year 1	15,500	\$42,034	\$524,148	\$550,355	\$646,551	\$3.4761
Year 2	16,275	\$43,675	\$544,610	\$571,841	\$661,620	\$3.3877
Year 3	17,088	\$45,382	\$565,892	\$594,187	\$687,474	\$3.3526
Year 4	17,943	\$47,157	\$588,025	\$617,426	\$714,362	\$3.3177
Year 5	18,840	\$49,003	\$611,043	\$641,595	\$742,325	\$3.2835

Discounted Annual Cost Trend



Formula Applied

Business Support	3.91% of Total Monthly Usage
Buffer	5% of Total Platform Annual Cost
Managed Services	30% of Total AWS Annual Cost
Discount	11% off Total Cost
PUPM	Discounted Annual Cost ÷ 12 ÷ Named Users
User Growth	+5% per year (Year-over-Year)
One-Time Costs (Y1)	Performance: \$5,000 · Migration: \$5,000 · Managed Setup: \$1,000

GCP Compute Engine Pricing

Google Cloud equivalent — US Central (Iowa)

Monthly	Annual	5-Year
\$19,468	\$233,620	\$1,265,360

Category	Role	Nodes	GCP Machine	\$/hr	\$/month	\$/year
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	6	n2-standard-16	\$1.17	\$5,378	\$64,540
Kubernetes Cluster	Worker Nodes Linux/RHEL for Grafana & Prometheus	1	n2-standard-8	\$0.58	\$513	\$6,162
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	n2-standard-8	\$0.58	\$1,201	\$14,412
PGSQL Database	Linux Nodes – Database server (Primary)	2	n2-highmem-32	\$3.86	\$5,734	\$68,804
PGSQL Database	Nodes – Cluster (etcd+haproxy, pgbackrest)	4	n2-highmem-2	\$0.24	\$772	\$9,263
PGSQL Database	Storage: SAN (Primary PostgreSQL DB)	2	Persistent Disk Extreme	\$0.00	\$525	\$6,300
S3	S3 Storage + Replication	1	Persistent Disk Extreme	\$0.00	\$738	\$8,850
Kubernetes Cluster	Cloud Managed K8s	1	GKE Standard Cluster	\$0.10	\$73	\$876
Caching & Session	Elasticache Service	2	Memorystore Redis (redis-16gb)	\$0.24	\$358	\$4,292
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	Cloud Load Balancing	\$0.00	\$35	\$420
Infrastructure & Monitoring	Bastion Host	1	n2-standard-2	\$0.15	\$107	\$1,279
Infrastructure & Monitoring	NAT Gateway	2	Cloud NAT	\$0.00	\$32	\$384
Infrastructure & Monitoring	Image Registry (ECR)	1	Artifact Registry	\$0.00	\$51	\$614
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage (Nearline)	\$0.00	\$205	\$2,458
ClickHouse DB Cluster	ClickHouse DB — Shard 1, Node 1	1	n2-highmem-8	\$0.96	\$1,454	\$17,447
ClickHouse DB Cluster	ClickHouse DB — Shard 1, Node 2	1	n2-highmem-8	\$0.96	\$1,454	\$17,447
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 1	1	n2-standard-4	\$0.29	\$247	\$2,967
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 2	1	n2-standard-4	\$0.29	\$247	\$2,967

Category	Role	Nodes	GCP Machine	\$/hr	\$/month	\$/year
ClickHouse Keeper Cluster	ClickHouse Keeper — Node 3	1	n2-standard-4	\$0.29	\$247	\$2,967
Operations	Cloud Operations (Logging + Monitoring)	0	Cloud Operations Suite	\$0.00	\$98	\$1,170

GCP 5-Year Forecast

Year	Multiplier	Monthly Cost	Annual Cost	Cumulative
Year 1	1.0000×	\$19,468	\$233,620	\$233,620
Year 2	1.0400×	\$20,247	\$242,965	\$476,584
Year 3	1.0816×	\$21,057	\$252,683	\$729,268
Year 4	1.1249×	\$21,899	\$262,791	\$992,058
Year 5	1.1699×	\$22,775	\$273,302	\$1,265,360

AWS vs GCP Cost Comparison

Side-by-side cost analysis — same workload, two clouds

\$12,390	\$19,468	\$7,078	AWS
AWS Monthly	GCP Monthly	Monthly Diff	5-Yr Winner
us-east-1	us-central1	AWS is cheaper	Saves \$460,035

AWS Annual	\$148,685
GCP Annual	\$233,620
AWS 5-Year Total	\$805,325
GCP 5-Year Total	\$1,265,360
Cheaper (Monthly)	AWS
Cheaper (5-Year)	AWS
5-Year Saving	\$460,035

Category-Level Breakdown

Category	AWS \$/mo	GCP \$/mo	Difference	% Diff	✓ Cheaper
Caching & Session	\$482	\$358	\$124	34.7%	GCP
ClickHouse DB Cluster	\$2,236	\$2,908	\$672	23.1%	AWS
ClickHouse Keeper Cluster	\$468	\$742	\$273	36.8%	AWS
Infrastructure & Monitoring	\$329	\$430	\$100	23.3%	AWS
Kubernetes Cluster	\$4,124	\$7,166	\$3,042	42.5%	AWS
Operations	\$98	\$98	\$0	0.0%	AWS
PGSQL Database	\$3,916	\$7,031	\$3,114	44.3%	AWS
S3	\$738	\$738	\$0	0.0%	AWS
Security & Compliance	\$0	\$0	\$0	0.0%	AWS

5-Year Year-by-Year Comparison

Year	AWS/mo	GCP/mo	AWS/yr	GCP/yr	AWS Cumul.	GCP Cumul.	Winner
Year 1	\$12,390	\$19,468	\$148,685	\$233,620	\$148,685	\$233,620	AWS
Year 2	\$12,886	\$20,247	\$154,632	\$242,965	\$303,317	\$476,584	AWS
Year 3	\$13,401	\$21,057	\$160,818	\$252,683	\$464,135	\$729,268	AWS
Year 4	\$13,938	\$21,899	\$167,250	\$262,791	\$631,385	\$992,058	AWS
Year 5	\$14,495	\$22,775	\$173,940	\$273,302	\$805,325	\$1,265,360	AWS

Notes & Assumptions

Detailed pricing basis, scope, and validity

■ Pricing Validity — 15 Days Only

This estimate is valid for **15 calendar days** from the date of issue (**09 June 2026**). AWS On-Demand prices, instance availability, and currency exchange rates are subject to change without notice. A revised estimate must be requested if requirements change or the validity period lapses.

Pricing & Infrastructure Assumptions

AWS Pricing Basis	All prices are AWS On-Demand, Linux/UNIX, shared tenancy. No Reserved Instances or Savings Plans applied. Switching to 1-Year Reserved pricing typically reduces compute costs by 30-40%.
Region	Estimate uses region: us-east-1. Costs vary by region — a regional multiplier is applied to the base us-east-1 rate.
Inflation Rate	4% per annum applied to all compute and managed-service costs from Year 2 onwards, reflecting typical AWS price adjustments and currency exposure.
High Availability (AZ)	Production and DR environments are architected as Multi-AZ (High Availability). Pre-Prod/SIT/UAT are Single-AZ to optimize costs while maintaining functional parity.
Security & Compliance Costs	Fixed INR-converted estimates included: Airtel SOC \$500/mo, SOC Machines Rs.30,000/yr, Req 4 Antivirus Rs.2,600/yr, Req 6 Data Discovery Rs.4,000/yr. These are not subject to AWS price changes.
Pre-Prod / SIT / UAT	Environments sized at ~40% of Production infrastructure. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server / Oracle use AWS Managed. Costs are additive and not included in the Production PUPM.
Disaster Recovery (DR)	DR sized using a Pilot Light / Warm Standby model at ~50% of Production compute. Active-Active DR would approximately double the DR cost shown.
Managed Services	Managed Services fees (30% of AWS annual total) cover platform monitoring, patching, incident management, and lifecycle support by the BusinessNext operations team.

Scope of This Estimate

Included	Production cloud infrastructure (compute, database, storage, networking)
Included	Pre-Production / SIT / UAT environments (if selected)
Included	Disaster Recovery environment (if selected)
Included	BusinessNext platform licensing and managed services
Included	One-time migration, performance testing, and managed-service setup (Year 1)
Excluded	Application development or customisation effort
Excluded	End-user device provisioning or on-site hardware
Excluded	Third-party SaaS subscriptions (e.g. Salesforce, Twilio, Okta)
Excluded	Data centre co-location or physical network charges
Excluded	Regulatory compliance certification fees

Legal Disclaimer

This document is prepared by BusinessNext for the exclusive use of the named client organisation. All cost figures are estimates based on current AWS/GCP published pricing and the inputs provided at the time of generation. Actual costs may vary depending on usage patterns, reserved-instance commitments, enterprise discount programmes, and any changes to cloud provider pricing.

This estimate does **not** constitute a binding commercial offer or contract. A formal Statement of Work (SOW) and Order Form must be executed prior to commencement of any services. BusinessNext reserves the right to revise this estimate upon receipt of updated requirements or upon expiry of the 15-day validity period.

Confidentiality: This document is CONFIDENTIAL. It must not be reproduced, distributed, or disclosed to any third party without prior written consent from BusinessNext.

Generated by BusinessNext Cost Estimator · 09 June 2026 · Internal Use Only · Pricing valid for 15 days from 09 June 2026